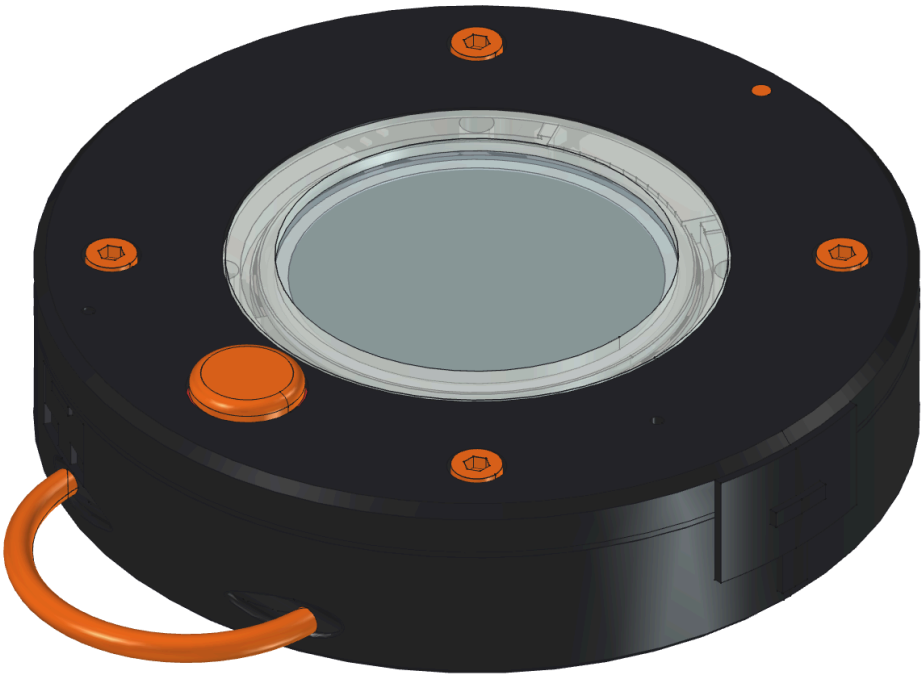
Finder Puck

RESEARCH + DESIRE

Build Guide

Round-screen puck, through-bolted · Version 0.2.0

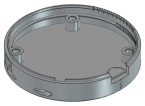


Support
support@researchanddesire.com

Design files
README.md in the finder-puck repository

Parts

3D printed



1×
Shell
black



1×
Plate
black



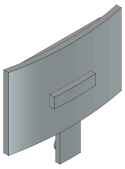
1×
Ring
black



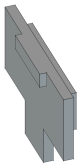
1×
Bezel
clear resin



1×
Button cap
orange

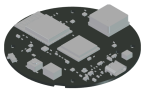


1×
USB-C plug
TPU 85A–95A, black



1×
Switch plug
TPU 85A–95A, black

Electronics



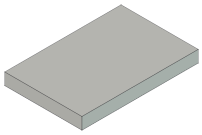
1×
Assembled board
factory-assembled PCB



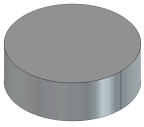
1×
Round display
1.28″ GC9A01, bare panel



1×
Crystal
Ø37.5 × 1.0 mm glass



1×
LiPo cell
503450, 28 AWG leads, JST PH



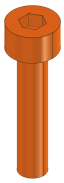
1×
Vibration motor
Ø10 mm coin, JST SH lead



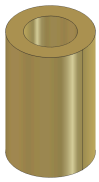
1×
915 MHz antenna
adhesive flex 40 × 7 mm (2J 2JF0515P), U.FL

Parts (continued)

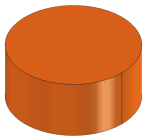
Hardware



4×
Bolt M2.5 × 12
socket cap, aluminium (non-magnetic)



4×
Heat-set insert M2.5
Ruthex RX-M2.5x5.7, 4 mm hole



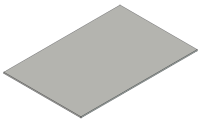
1×
Index stub
orange 1.75 mm filament, trim flush



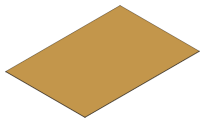
1×
Foam ring
0.5 mm, Ø36.3–38 mm, open 10 to 5 o'clock



1×
Gasket
0.3 mm foam, Ø32.5–35 mm



1×
Battery pad
0.5 mm foam, 34 × 50 mm



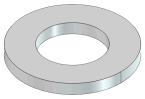
1×
Insulation sheet
0.2 mm Kapton or PET, 36 × 52 mm



1×
Cord
Ø2.5 mm, 300 mm



1×
Seam seal
1.0 mm closed-cell foam, Ø73.5–76.3 mm, open at 3 o'clock



4×
Washer M2.5
nylon DIN 433, 5 × 2.7 × 0.5 mm



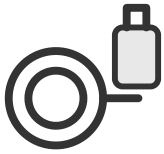
1×
O-ring
9 × 1 mm silicone

Tools

Required



Soldering iron, fine tip



Solder and flux



Tweezers



Hex key 2 mm



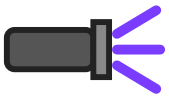
Multimeter



Super glue



USB-C cable and a computer



Clear UV resin and a UV torch

Recommended



Magnifier



Isopropyl alcohol



Kapton tape



Paint pen



Silicone grease

Before you start

Print

Shell upright, ring upside down (top face on the bed), plate face down, cap pocket down; 0.2 mm layers, four walls. Print the bezel in clear resin: its flange cannot be bridged and its fits are 0.25 mm. Print the two plugs in TPU, standing on their strap ends. Print a hole coupon before trusting the insert holes.

Sealing

The puck is built to IP55 (dust, water jets). That holds only with every seal in: the seam seal, a washer under each bolt head, the O-ring on the cap, UV resin round the crystal and the bezel, resin in the pinhole and both plugs pushed fully home. Cut the seam seal with a compass cutter.

Panel first

Buy two panels. Before anything else, look at the flex: after it folds under the glass, pin 1 must land on the 3 o'clock side of the board's pads. If yours is mirrored, the board needs regenerating with FPC_PIN1_RIGHT flipped, not a different fold.

Battery

The cell is live whatever the switch does. It is plugged in only after the soldering (step 9), and the insulation sheet goes on it before the board. Check the lead's polarity with the meter; many cells are wired the other way round.

Iron

220 °C for the inserts, 320 °C with plenty of flux for the 0.7 mm flex. Clean with alcohol.

Compass

Keep steel away from the compass: aluminium bolts only, no steel screws or magnets near the puck.

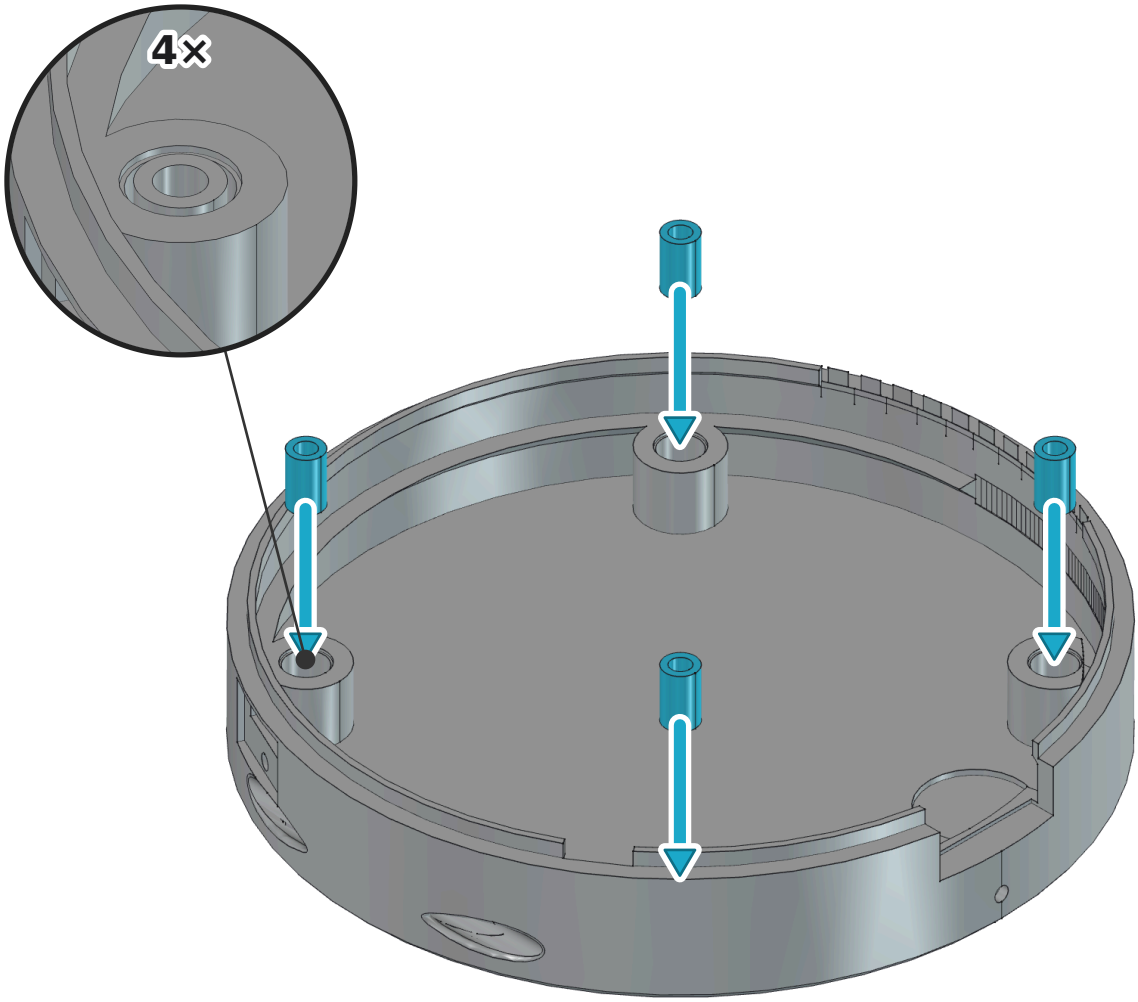
1



1×
Shell

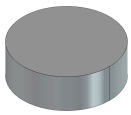


4×
Heat-set insert M2.5



Heat-set with the iron at 220 °C until the insert is flush with the boss.

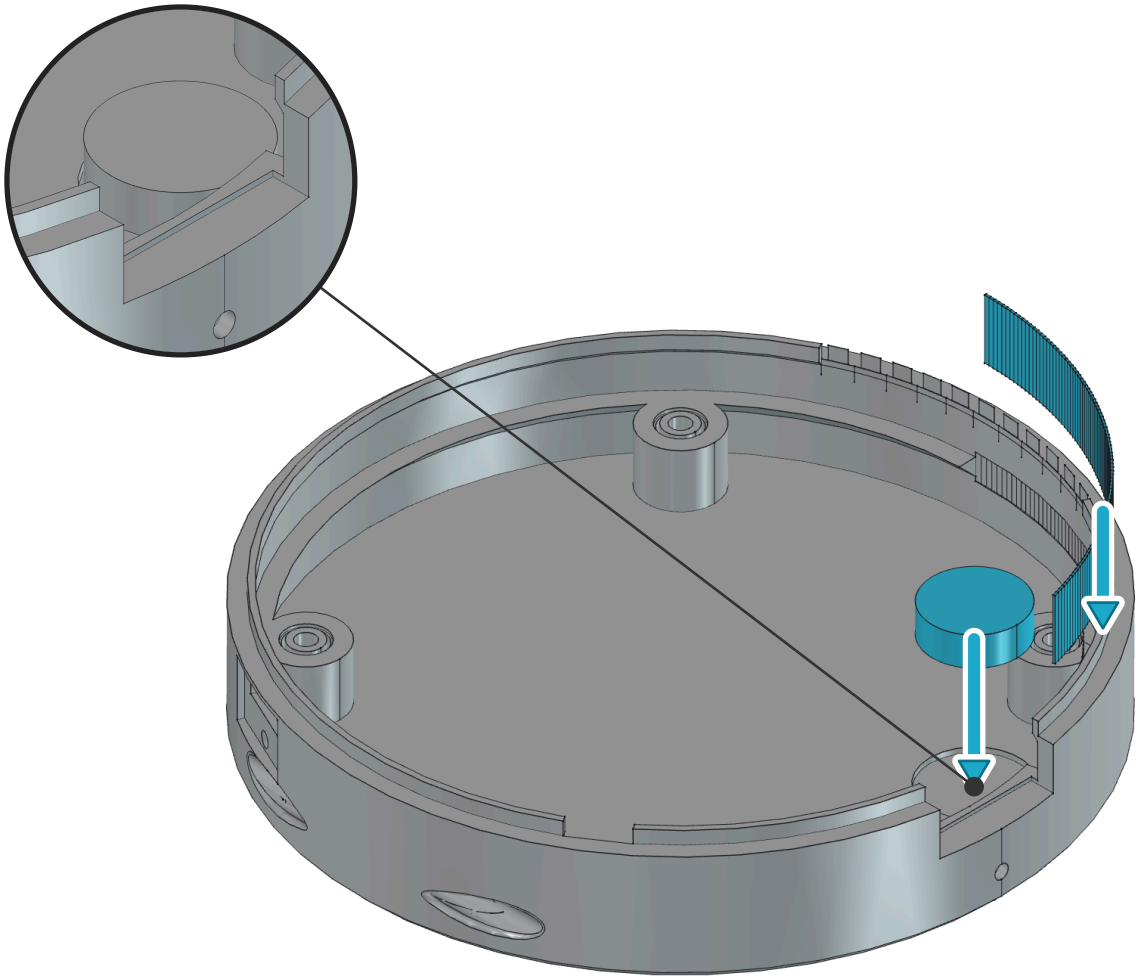
2



1×
Vibration motor

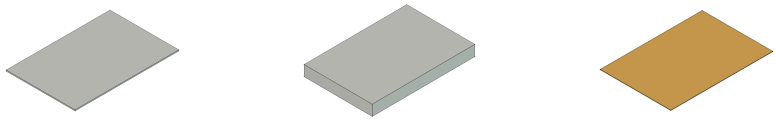


1×
915 MHz antenna



Motor in the pocket at 2 o'clock, lead up. Peel the flex antenna and press it to the inside of the wall between 12 and 3 o'clock, cable end at 3 o'clock; run the pigtail along the floor to the notch at 5 o'clock.

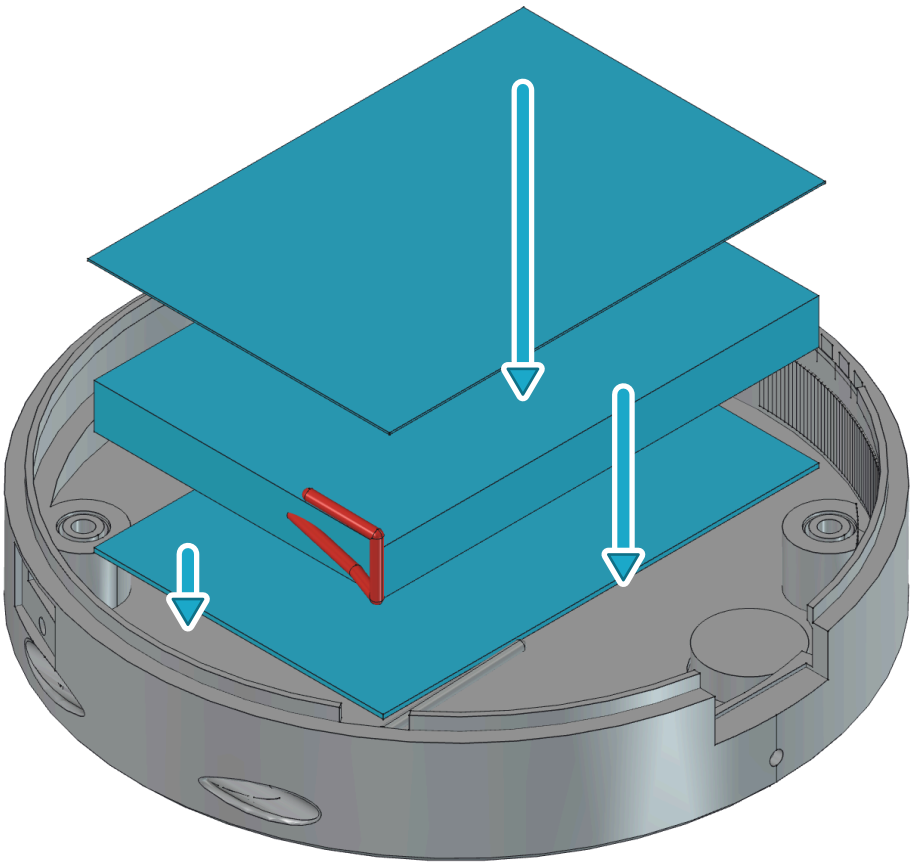
3



1× Battery pad

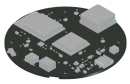
1× LiPo cell

1× Insulation sheet

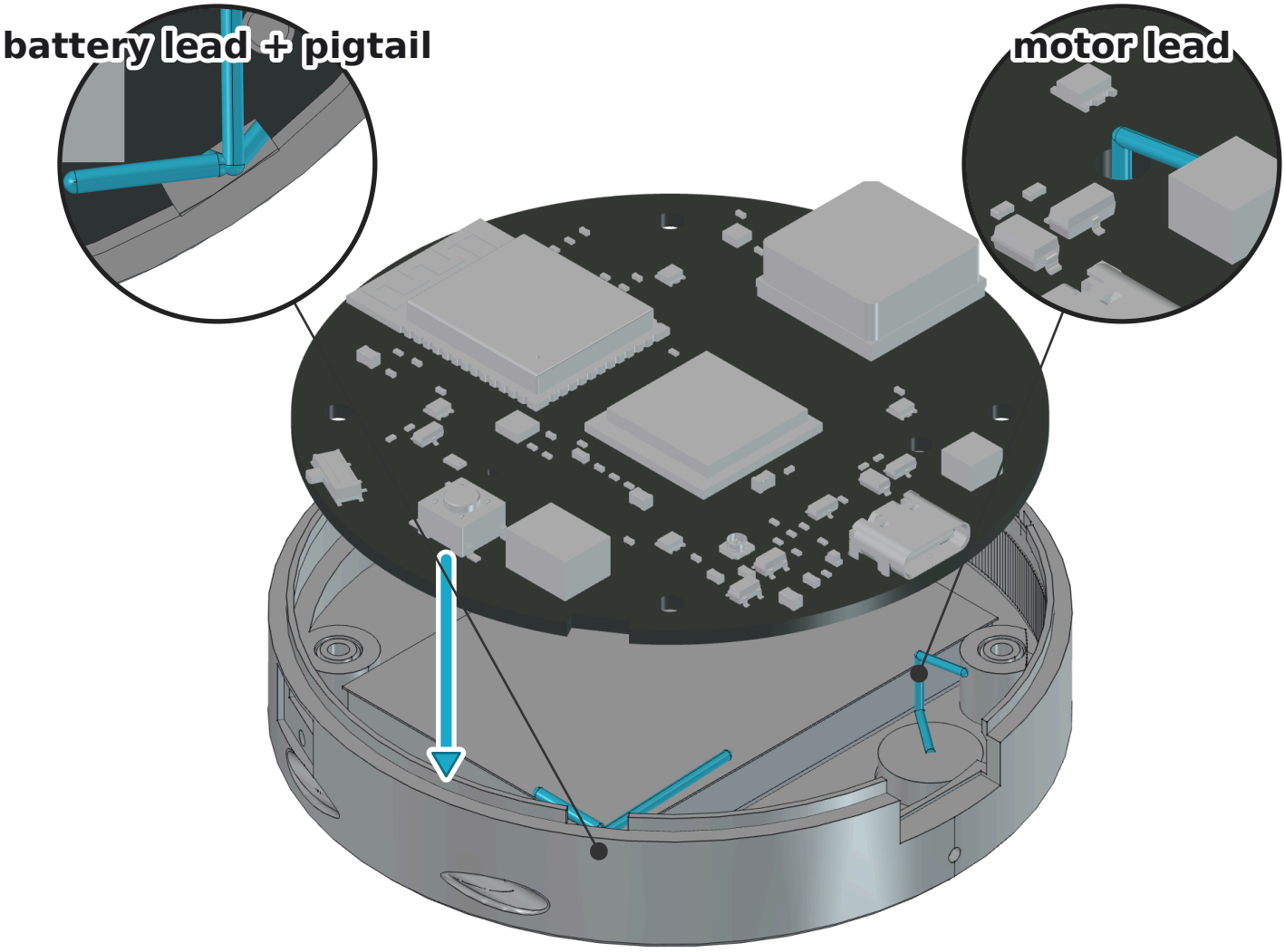


Pad first, the cell on it with its lead toward 5 o'clock, then the insulation sheet on top of the cell. Do not pinch the lead.

4

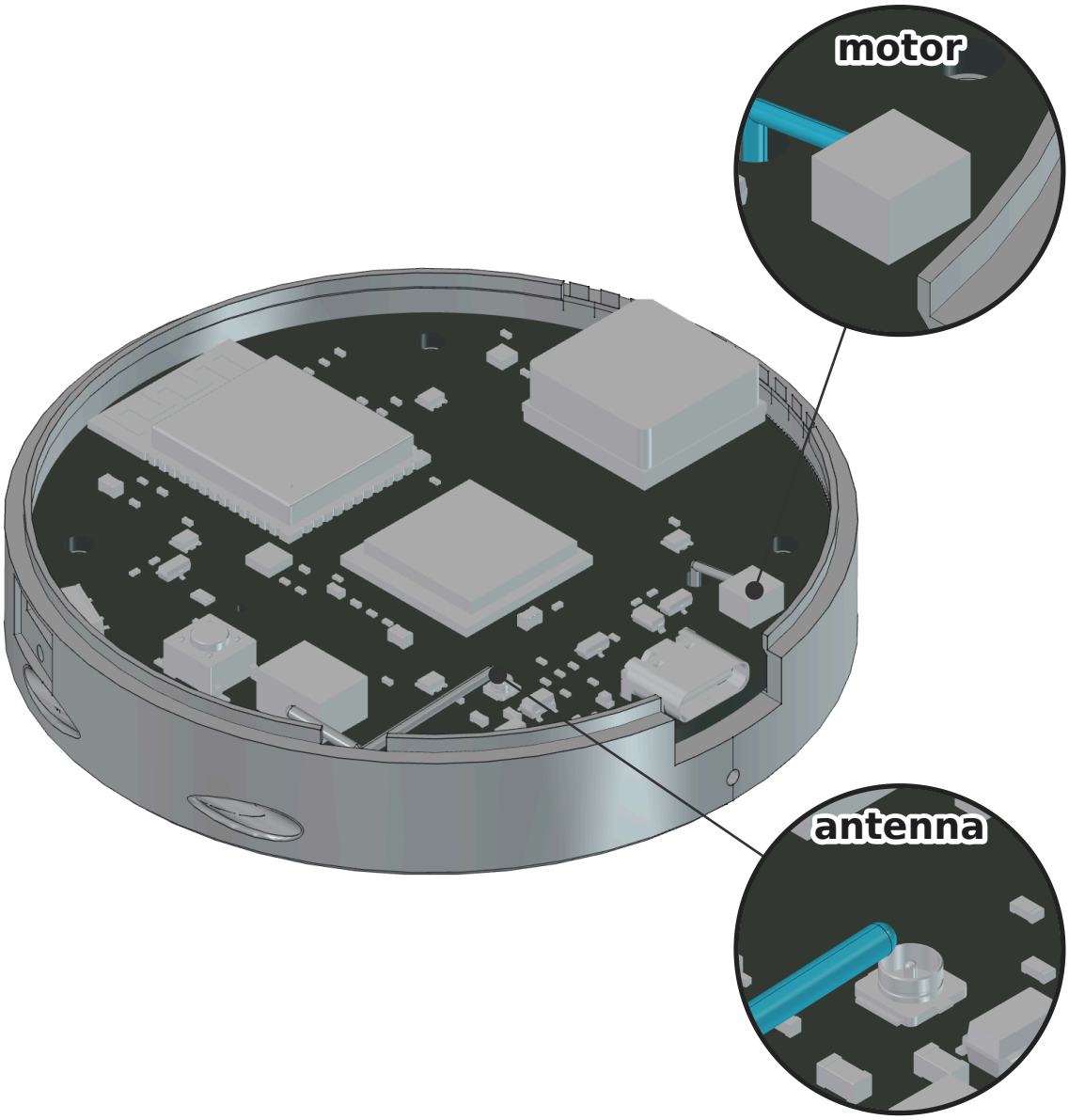


1×
Assembled board



Feed the motor lead up through its hole, the battery lead and the antenna pigtail up through the 6 mm notch at 5 o'clock, then lower the board onto the rim.

5



Plug the motor into its socket and click the pigtail onto the antenna socket. The battery stays unplugged until the panel is soldered.

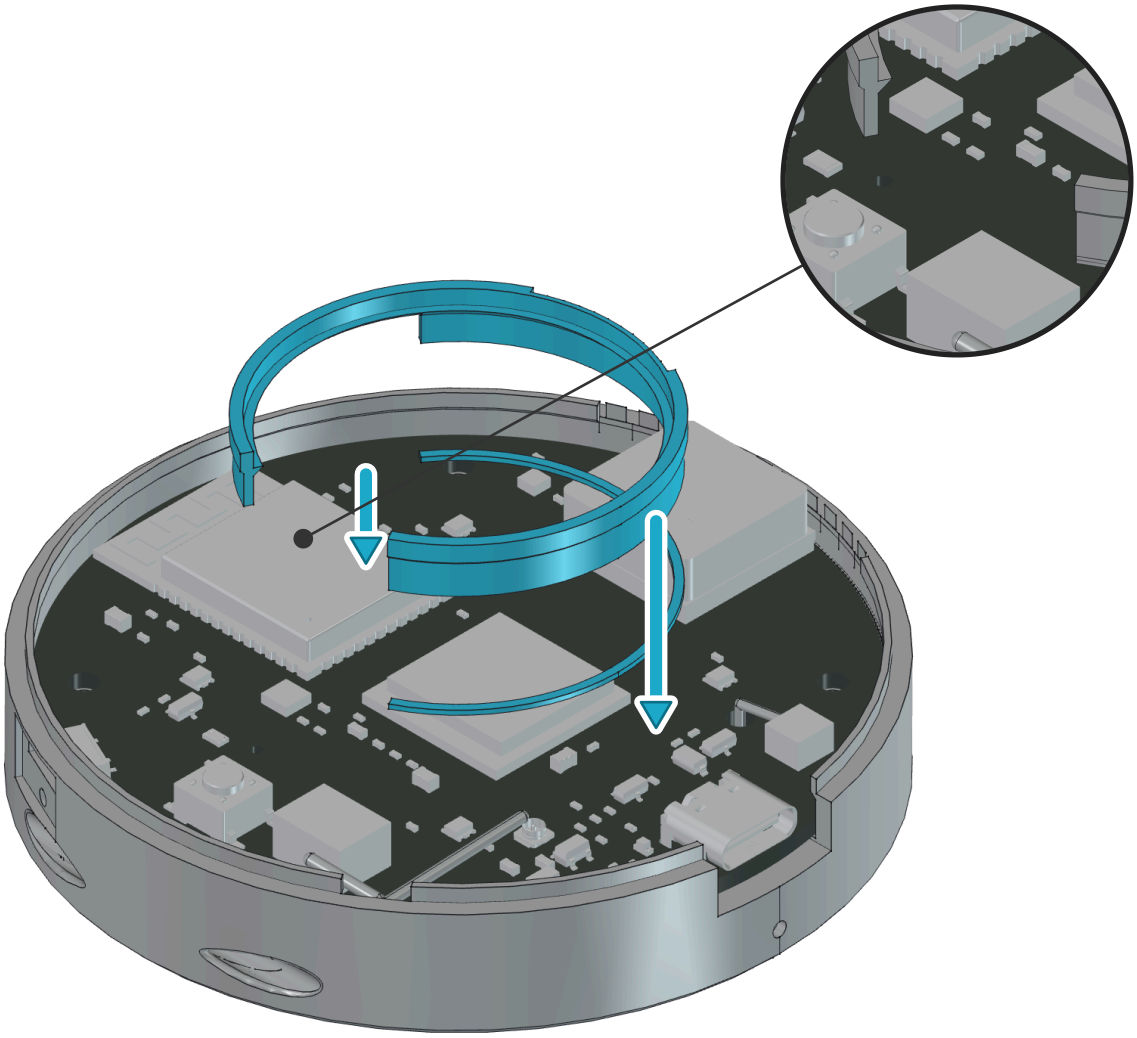
6



1×
Foam ring



1×
Ring

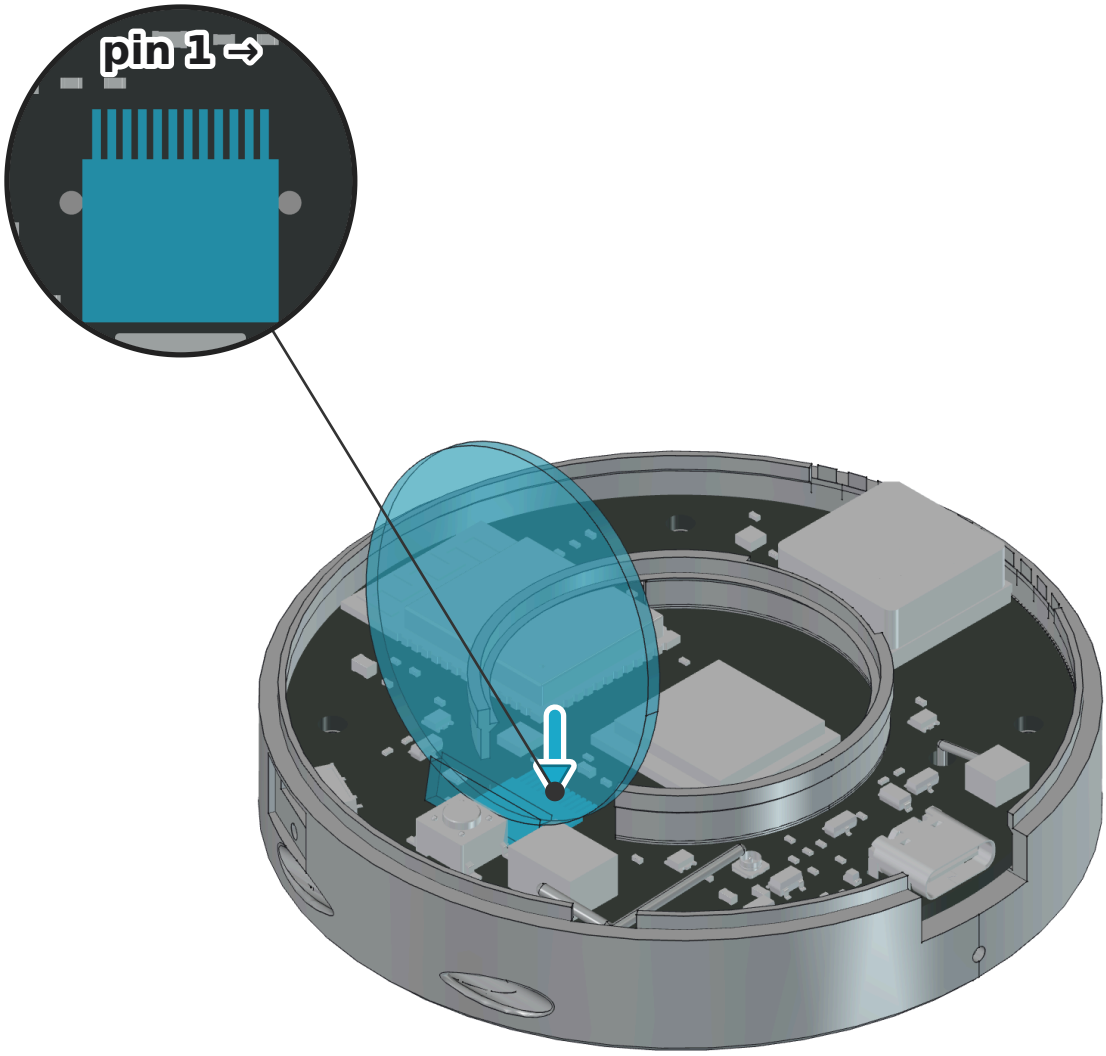


The opening in the ring faces the button (6 o'clock); the solder pads stay reachable through it.

7

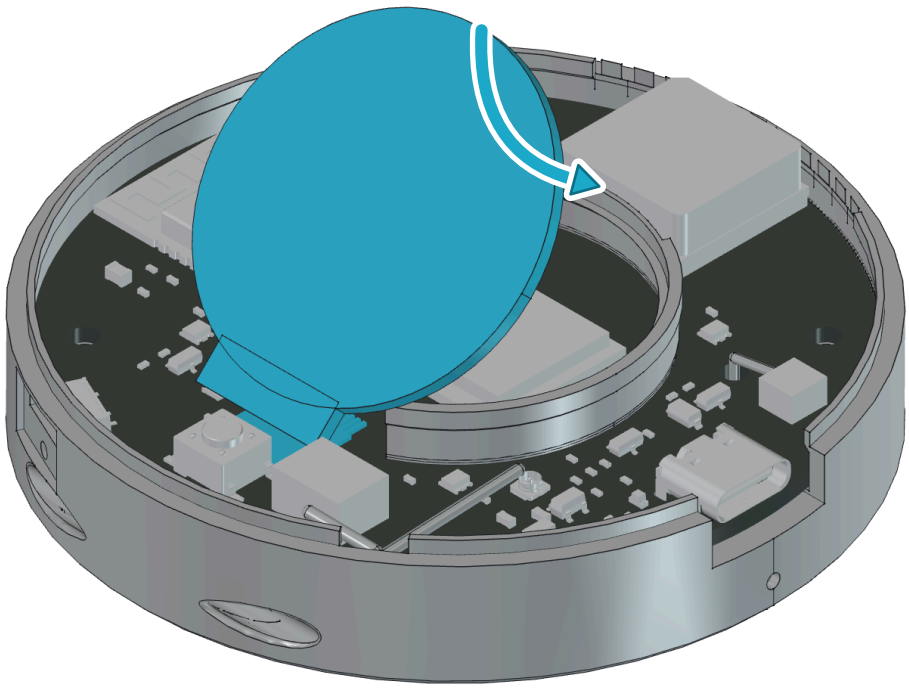


1×
Round display



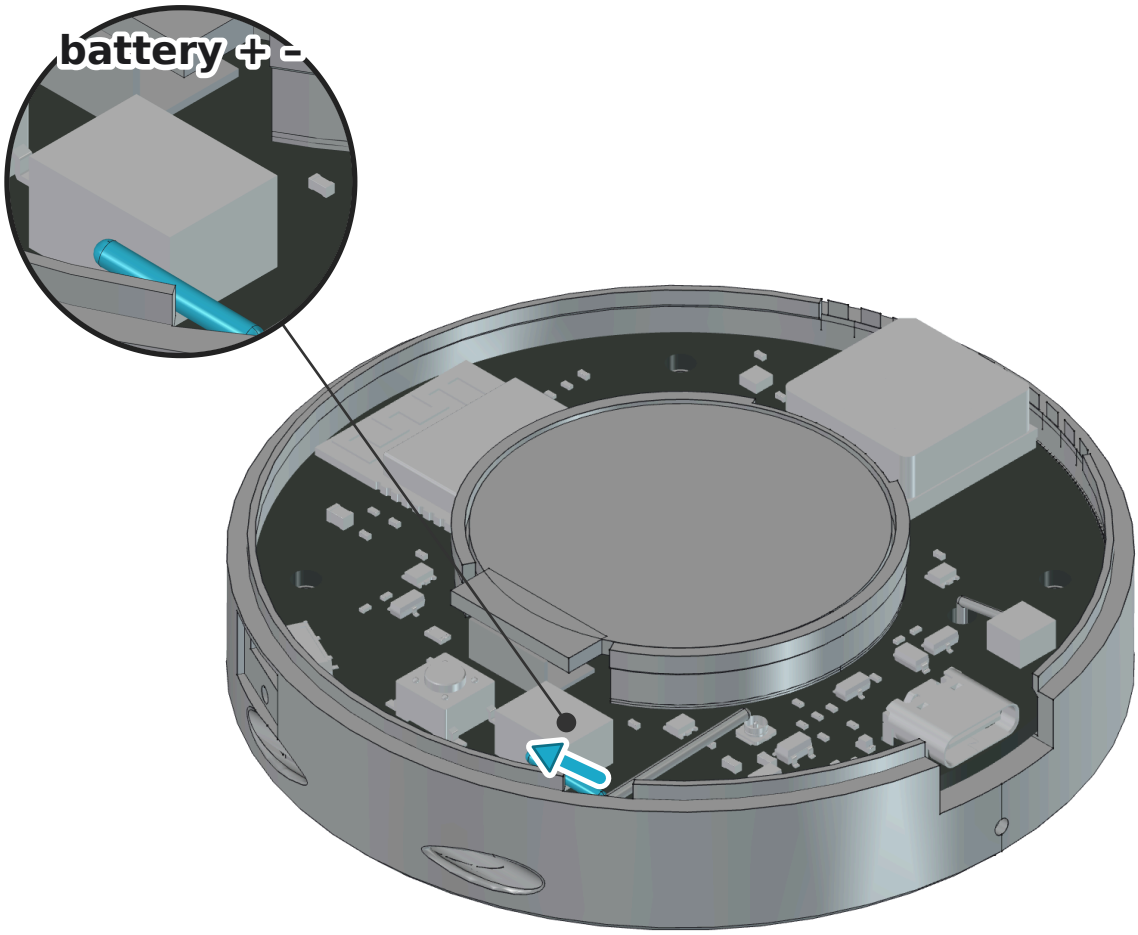
Stand the panel on its chin and lay the flex on the pads: pin 1 on the right as you look from the button side (use the loupe). Tack both ends, solder the 12 fingers, clean with alcohol.

8



Fold the panel down onto the ledge. The flex bends once, under the glass; never crease it.

9

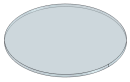


Now plug in the battery. Check the lead with the meter first: + is marked on the board next to the socket.

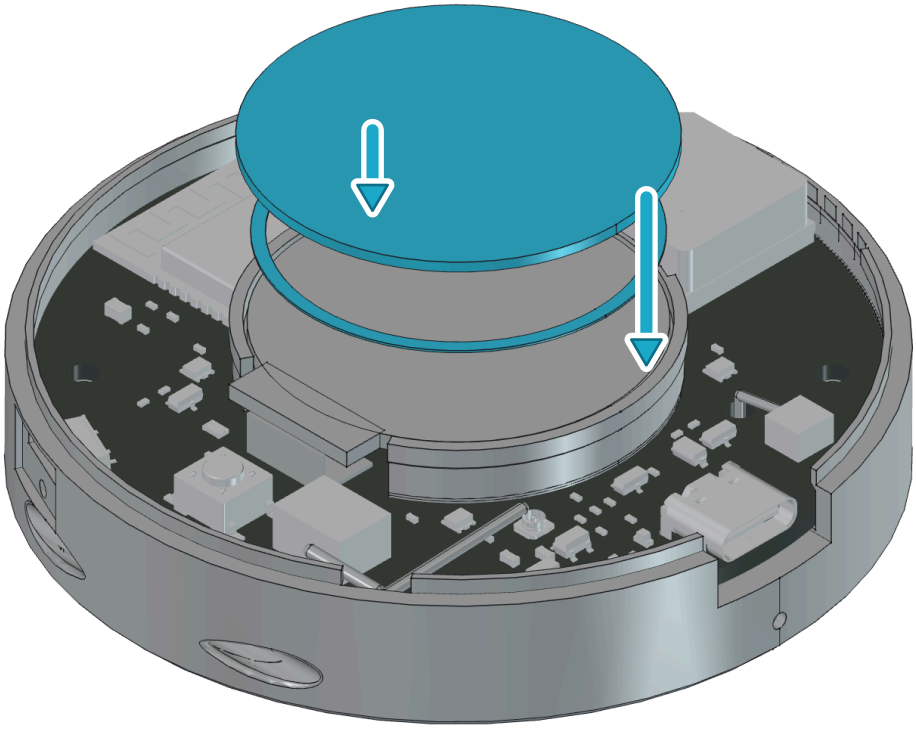
10



1×
Gasket



1×
Crystal

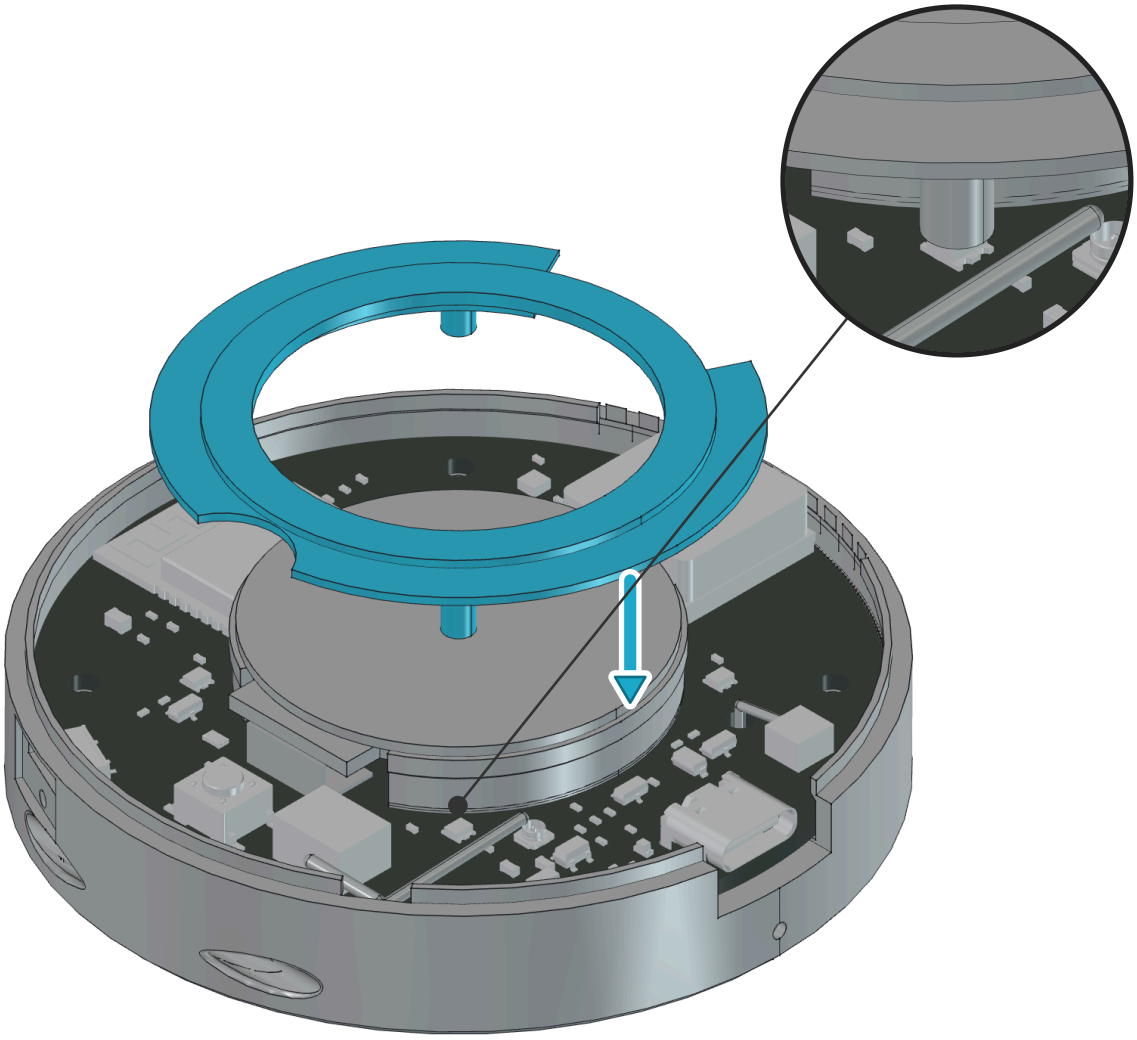


Blow the dust off the glass first. The crystal rests on the ring, not on the panel. Run a thin bead of UV resin round the crystal's top edge, but do not cure it yet.

11



1×
Bezel



The four light pipes drop over the LEDs; the bezel seats over the ring and holds the crystal. Cure the resin through the clear lip with the UV torch.

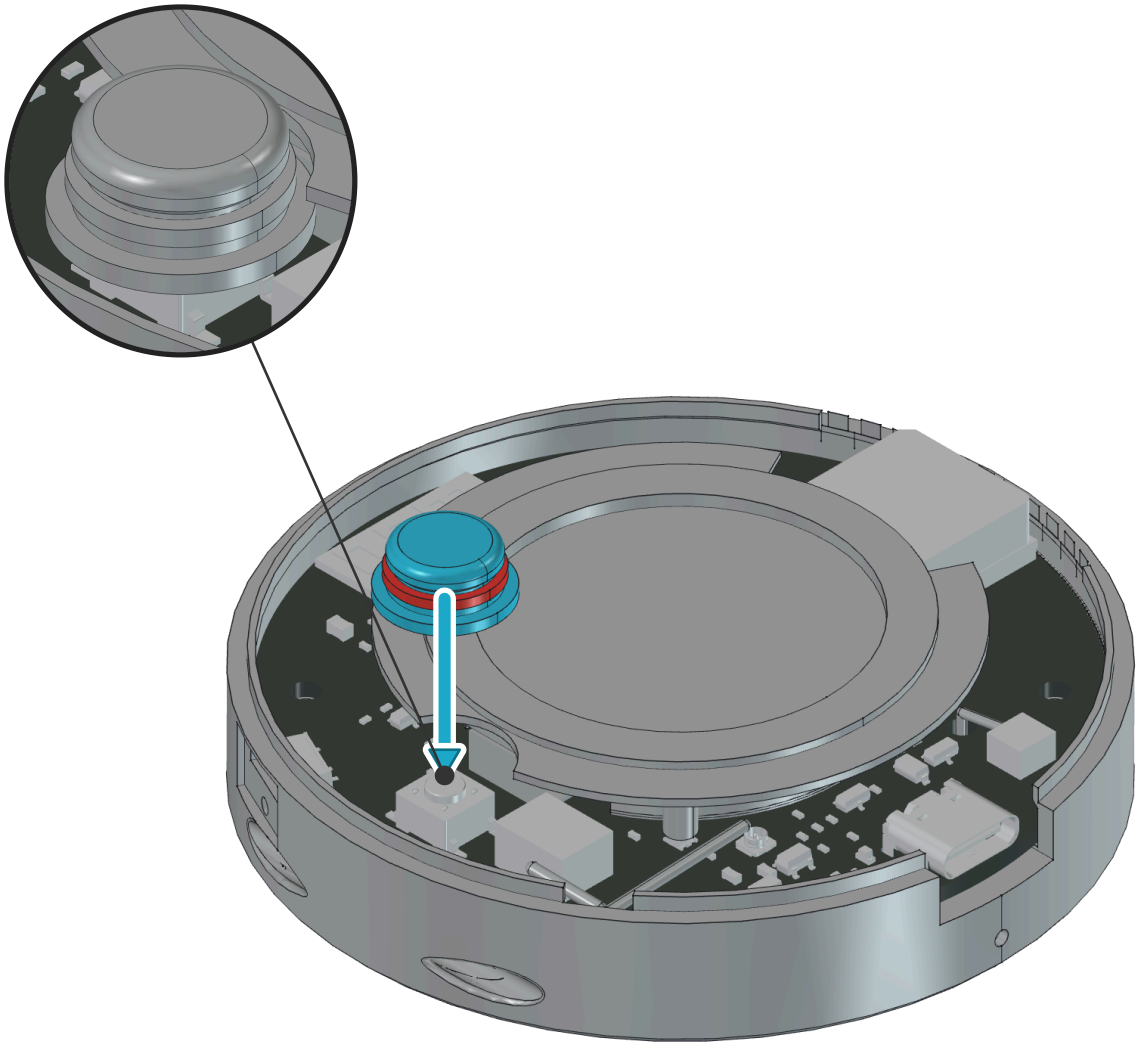
12



1×
Button cap

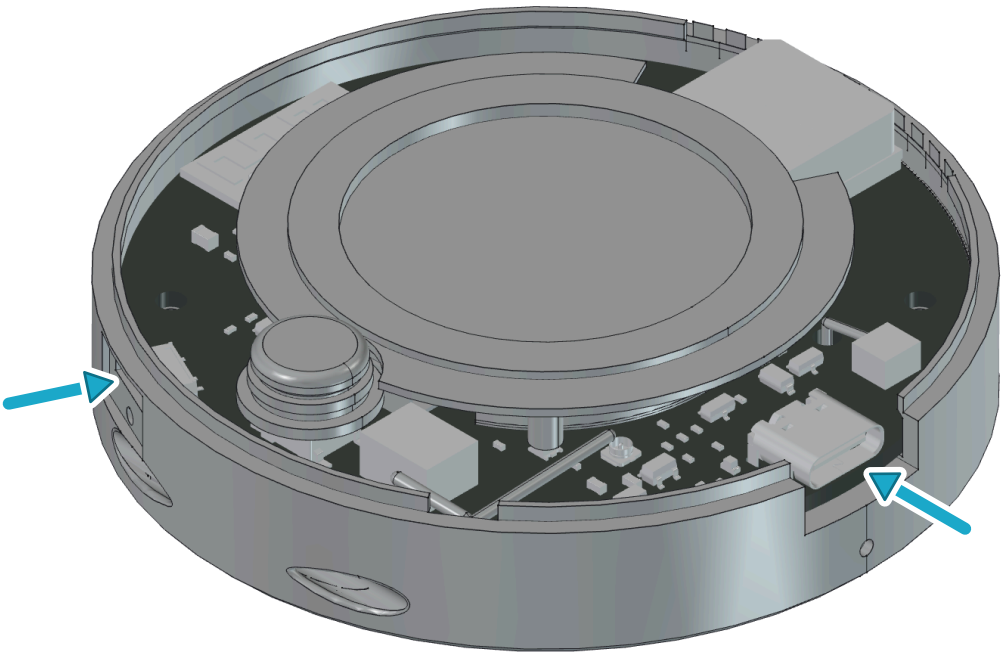


1×
O-ring



Roll the O-ring into the cap's groove and grease it. The square pocket in the cap fits the switch stem; a dot of glue in the pocket keeps it on.

13



Test before the plate goes on: plug in USB-C, flash the firmware, slide the switch toward the dot. The screen lights, the four LEDs strobe when you press the button, the motor ticks. Switch off again.

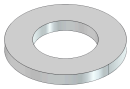
14



1×
Seam seal



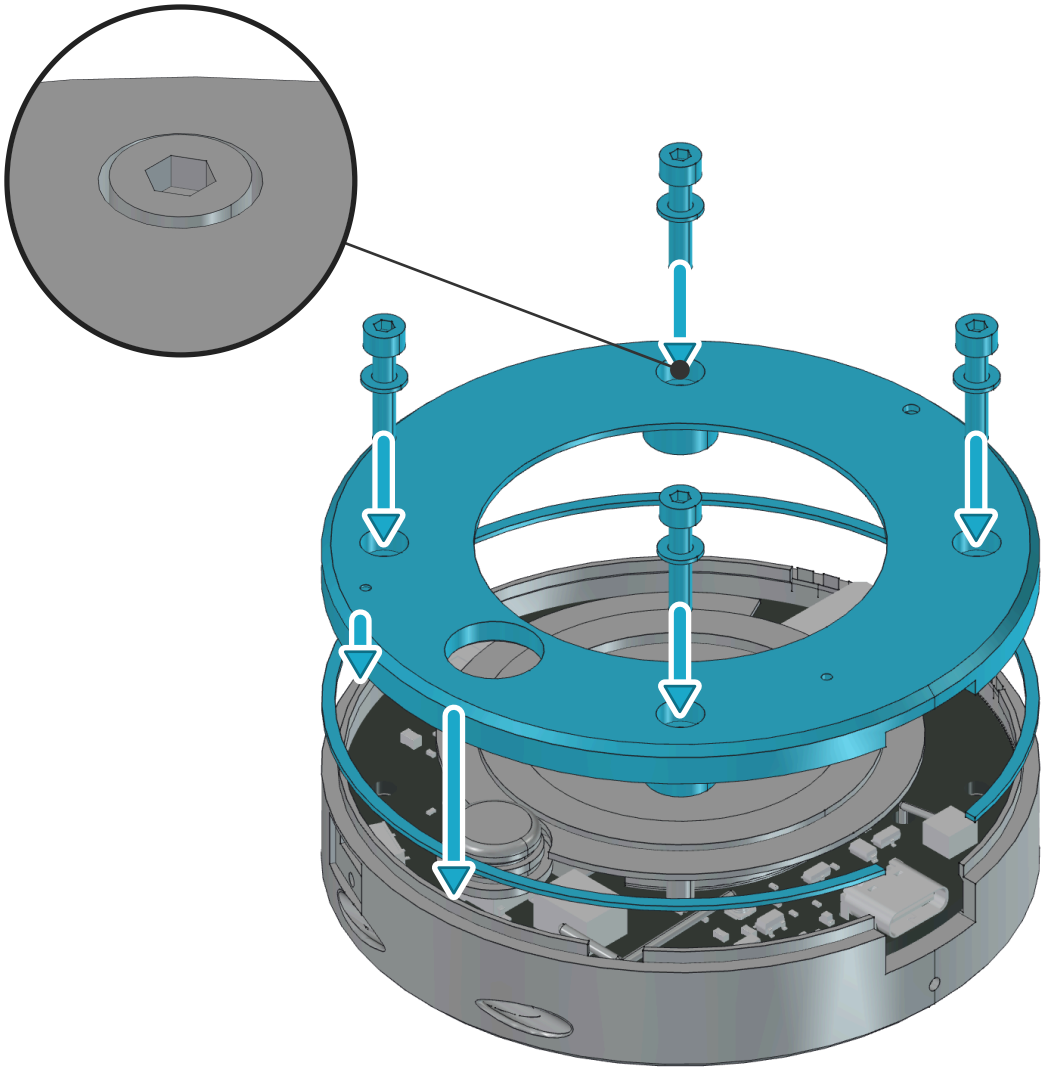
1×
Plate



4×
Washer M2.5

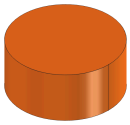


4×
Bolt M2.5 × 12



Stick the seal on the shell rim, its ends against the USB-C opening. Run UV resin round the bezel where it meets the plate, lower the plate over the bezel and cap, then the bolts, each on a washer. Hand-tight only, in a cross: aluminium sockets strip. Cure through the lip.

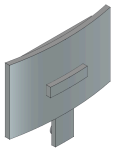
15



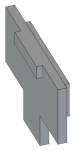
1×
Index stub



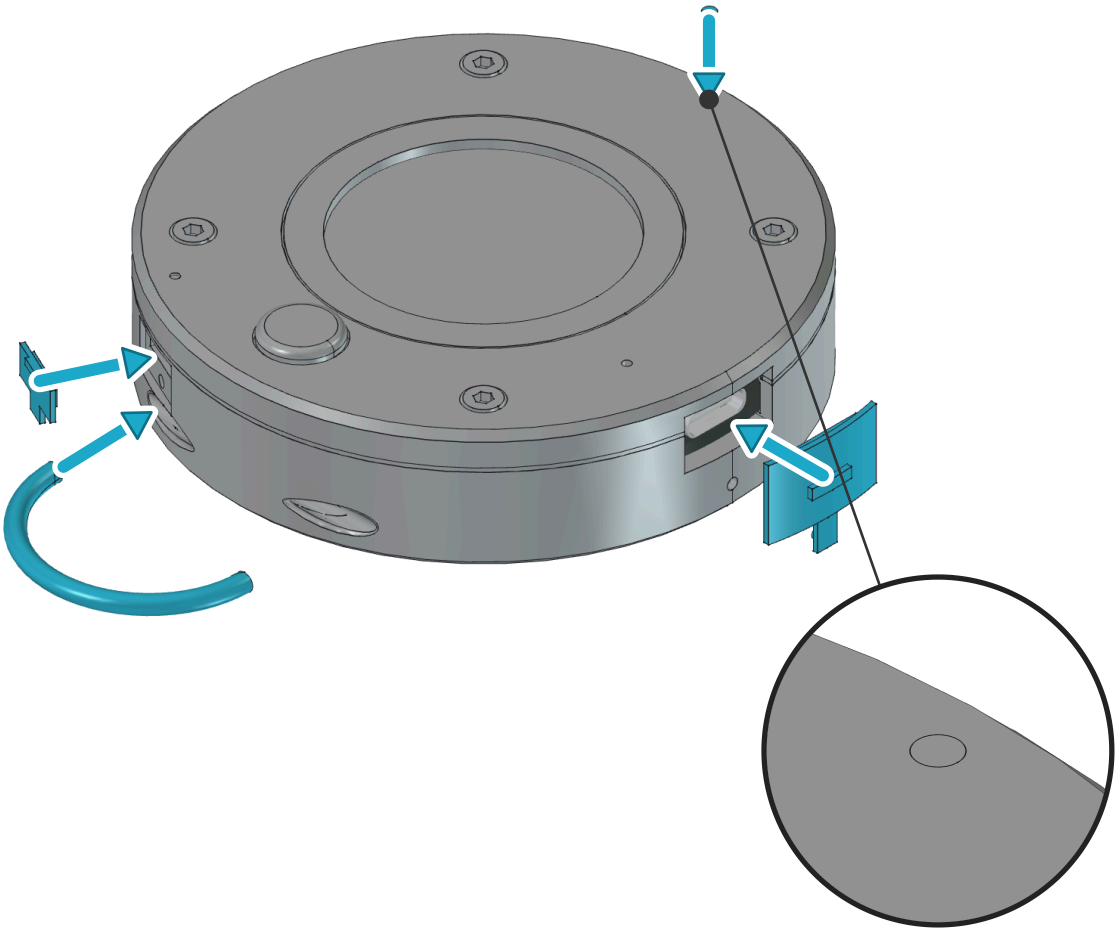
1×
Cord



1×
USB-C plug

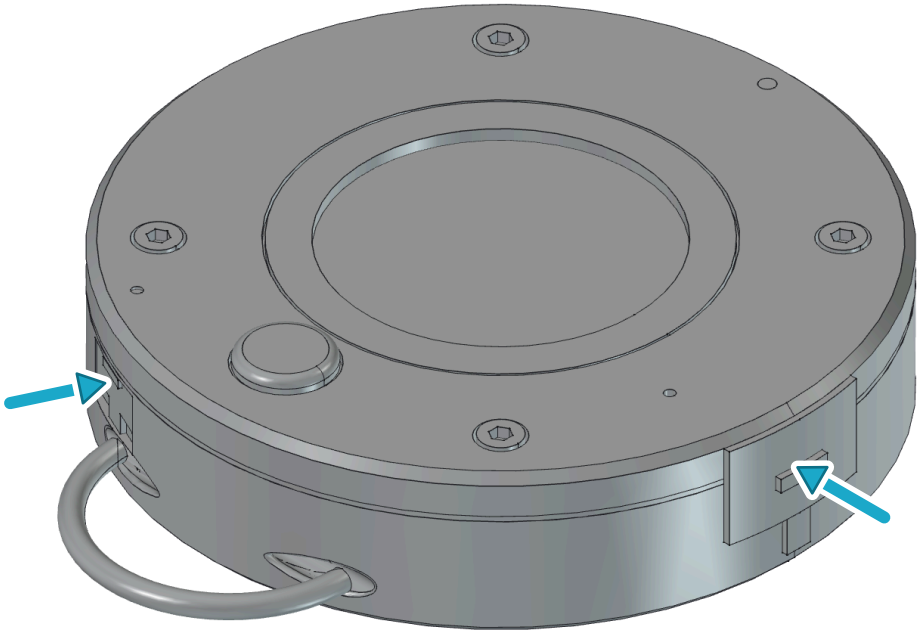


1×
Switch plug



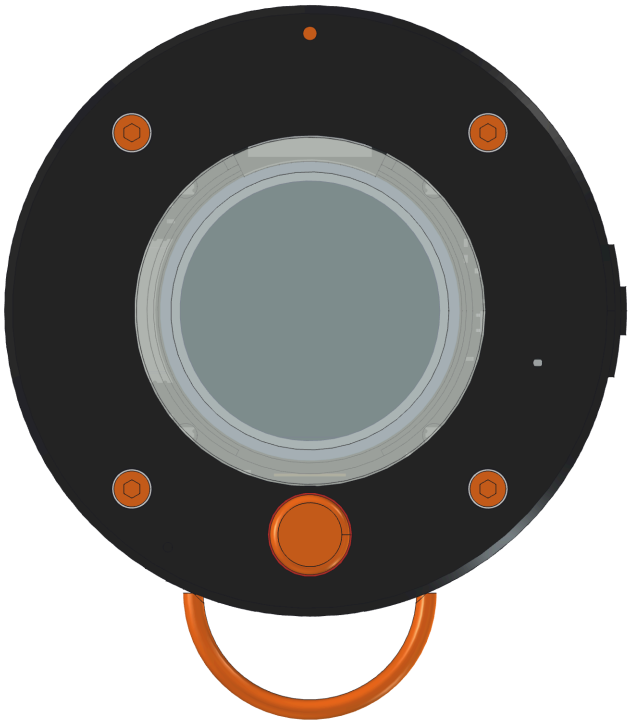
Glue the orange stub into the hole at 12 o'clock all round and trim it flush; fill the pinhole flush with UV resin and cure it. Glue each plug's pin into the hole under its opening and push the plugs home. Thread the cord in one mouth and out the other, then knot it.

16



Lift the switch plug and slide the switch toward the dot to switch on. Charge through USB-C with the puck off and dry; the pinhole LED goes out when the cell is full. Push both plugs fully home before the puck gets wet. Mark it 1 or 2 on the back with the paint pen.

Done



The orange dot is 12 o'clock. The arrow points at your partner and the blip sits at their range on the rings; the scale (50 m, 500 m, 5 km) is printed at 3 o'clock. While you hold the button the screen tells you what letting go will do.

The button

Press	Beacon on or off for 60 s: both pucks strobe so your friend can spot you.
Double press	Vibration on or off (remembered).
Hold 2 s, release	Pairing mode. Hold again to cancel.
Hold 5 s, release	Compass calibration: turn the puck through a slow figure of eight.

RESEARCH + DESIRE

Finder Puck Build Guide

Round-screen puck, through-bolted · Version 0.2.0

Support
support@researchanddesire.com

Design files
README.md in the finder-puck repository

Open hardware. The design files (KiCad board, FreeCAD case, this guide) are generated from scripts; see the README before changing anything by hand.